Apache Spark + Kafka with Log4j Logging

Setup Steps

- Create a Kafka cluster on any cloud provider.
- For this walkthrough, SASL is used for connecting to the Kafka server.
- Create a simple project directory that sends a DataFrame to Kafka.
- Use `log4j2.properties` for logging configuration.
- Create `local` , `qa` , and `prod` users for Kafka.
- Add these credentials to `conf/sbdl.conf` .
- Download and add the CA certificate reference to `conf/sbdl.conf` .
- Add the certificate to the machine and generate a `truststore.jks` file:

```
keytool -importcert \
  -alias my-ca \
  -file ca-certificate.crt \
  -keystore truststore.jks \
  -storepass changeit \
  -noprompt
```

- Add the certificate path and truststore location to `conf/sbdl.conf` .

Understanding `log4j2.properties`

1. Logger

- The **entry point** for logging.
- Defines what level of logs should be captured.

2. Appenders

- The **exit point** for logging.
- Determines where logs are written.

3. Important Types of Appenders

- `ConsoleAppender`
- `FileAppender`
- `RollingFileAppender`
- `MessageQueueAppender`
- `NetworkAppender`
- `DatabaseAppender`

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4. Log Levels

Name	Priority
OFF	0
FATAL	100
ERROR	200
WARN	300
INFO	400
DEBUG	500
TRACE	600
ALL	Integer.MAX_VALUE

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5. Log Filtering

- Log4j allows logging at different levels depending on the requirement.
- `Filters` can be configured to record only specific log events.

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6. Log Organization

- Different log levels can be written to different files.

- Helps keep logs clean and organized.

- Common practice:

  - `INFO` logs → `application.log`
  - `ERROR` logs → `error.log`
  - `DEBUG` logs → `debug.log`

7. References

[Log4j docs]